

On equidistribution in Arakelov theory

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Motivation: Bilu's Equidistribution Theorem

We define the *canonical height* of a non-zero integer polynomial $P(X) = a_n \prod_{j=1}^n (X - \alpha_j)$ (or of its zeros α_j 's if P is irreducible) to be

$$h_{\text{can}}(P) = \frac{1}{n} \log |a_n| + \frac{1}{n} \sum_{j=1}^n \log \max\{1, |\alpha_j|\}.$$

Theorem (Bilu)

Let $(P_n)_{n \geq 0}$ be a sequence of non-zero integer Polynomials with $\lim_{n \rightarrow \infty} h_{\text{can}}(P_n) = 0$, such that each P_n has pairwise different zeros, and μ_{S^1} the normed Haar measure on $S^1 \subseteq \mathbb{C}$. For any bounded, continuous function $f: \mathbb{C} \rightarrow \mathbb{C}$ it holds

$$\lim_{n \rightarrow \infty} \frac{1}{\deg P_n} \sum_{\substack{z \in \mathbb{C} \\ P_n(z)=0}} f(z) = \int_{S^1} f d\mu_{S^1}.$$

Questions:

- What happens if $h_{\text{can}}(P_n)$ does not tend to 0?
- Can we get equidistribution on whole $\mathbb{C}$ instead of S^1 ?

The Fubini–Study metric on the projective line

- $\mathbb{P}_{\mathbb{Z}}^1$: Projective line defined over $\mathbb{Z}$.
- Z_0, Z_1 : Homogeneous coordinates on $\mathbb{P}_{\mathbb{Z}}^1$.
- $H^0(\mathbb{P}_{\mathbb{Z}}^1, \mathcal{O}(n)) = \left\{ \sum_{j=0}^n a_j Z_0^j Z_1^{n-j} \mid a_j \in \mathbb{Z} \right\} \cong \{P \in \mathbb{Z}[X] \mid \deg P \leq n\}$
- $H^0(\mathbb{P}_{\mathbb{Z}}^1, \mathcal{O}(n)) \otimes_{\mathbb{Z}} \mathbb{C} \cong H^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{O}(n)) \cong \{P \in \mathbb{C}[X] \mid \deg P \leq n\}$
- *Fubini–Study metric*: For $s = \sum_{j=0}^n a_j Z_0^j Z_1^{n-j} \in H^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{O}(n))$ we define

$$|s|_{\text{FS}}^2: \mathbb{P}_{\mathbb{C}}^1 \rightarrow \mathbb{R}_{\geq 0}, \quad |s|_{\text{FS}}^2 = \frac{\left| \sum_{j=0}^n a_j Z_0^j Z_1^{n-j} \right|^2}{(|Z_0|^2 + |Z_1|^2)^n}.$$

- The curvature of the Fubini–Study metric on $\mathcal{O}(1)$ is given for the coordinate $z = Z_0/Z_1$ on $\mathbb{C} \cong \{Z_1 \neq 0\} \subseteq \mathbb{P}_{\mathbb{C}}^1$ by

$$c_1(\mathcal{O}(1)) = \frac{\partial \bar{\partial}}{2\pi i} \log |Z_1|_{\text{FS}}^2 = \frac{i}{2\pi} \frac{1}{(|z|^2 + 1)^2} dz \wedge d\bar{z} > 0.$$

Heights as integrals

- To a non-zero section $s \in H^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{O}(1))$ we associate the *Fubini-Study height* $h_{\text{FS}}(s)$ and the *Bombieri height* $h_{\text{B}}(s)$ by the integrals

$$h_{\text{FS}}(s) = \frac{1}{n} \int_{\mathbb{P}_{\mathbb{C}}^1} \log |s|_{\text{FS}} c_1(\mathcal{O}(1)) + \frac{1}{2}, \quad h_{\text{B}}(s) = \frac{1}{2n} \log \int_{\mathbb{P}_{\mathbb{C}}^1} |s|_{\text{FS}}^2 c_1(\mathcal{O}(1)).$$

- We identify sections with polynomials by the bijection

$$\{P \in \mathbb{C}[X] \mid \deg P \leq n\} \rightarrow H^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{O}(n))$$

$$P = \sum_{j=0}^n a_j X^j \mapsto s_P = \sum_{j=0}^n a_j Z_0^j Z_1^{n-j}.$$

- We set

$$h_{\text{FS}}(P) = h_{\text{FS}}(s_P) \quad \text{and} \quad h_{\text{B}}(P) = h_{\text{B}}(s_P).$$

Explicit formulas

- If $P = \sum_{j=0}^n a_j X^j = a_n \prod_{j=1}^n (X - \alpha_j)$ is a non-zero integer polynomial, a direct computation shows

$$h_{\text{FS}}(P) = \frac{1}{n} \log |a_n| + \frac{1}{2n} \sum_{j=1}^n \log(1 + |\alpha_j|^2), \quad h_{\text{B}}(P) = \frac{1}{2n} \log \left(\frac{1}{n+1} \sum_{j=0}^n \frac{|a_j|^2}{\binom{n}{j}} \right).$$

- More generally, for $p \geq 1$ we define

$$h_{\text{B}}^p(P) = \frac{1}{pn} \log \left(\frac{1}{n+1} \sum_{j=0}^n \frac{|a_j|^p}{\binom{n}{j}^{p/2}} \right), \quad h_{\text{B}}^{\infty}(P) = \frac{1}{n} \log \max_{j=0}^n \frac{|a_j|}{\sqrt{\binom{n}{j}}}.$$

- By Jensen's inequality ($\int \log \leq \log \int$) we get for any $p \in [2, \infty]$ that

$$h_{\text{FS}}(P) - \frac{1}{2} \leq h_{\text{B}}(P) \leq h_{\text{B}}^p(P).$$

Equidistribution of zero sets of integer polynomials

Theorem (W. 2022)

Let $p \in [1, \infty]$ and $f: \mathbb{C} \rightarrow \mathbb{C}$ be any continuous function, such that $\lim_{|z| \rightarrow \infty} f(z)$ is well-defined and finite.

- (a) Let $(P_n)_{n \geq 1}$ be a sequence of non-zero complex polynomials such that $\deg P_n = n$ and $\lim_{n \rightarrow \infty} (h_B^p(P_n) - h_{FS}(P_n) + \frac{1}{2}) = 0$. Then it holds

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{\substack{z \in \mathbb{C} \\ P_n(z)=0}} f(z) = \frac{i}{2\pi} \int_{\mathbb{C}} f(z) \frac{dz \wedge d\bar{z}}{(|z|^2+1)^2}.$$

- (b) Let $(r_n)_{n \geq 1}$ be a sequence of reals satisfying $\liminf_{n \rightarrow \infty} r_n > 0$ and $\limsup_{n \rightarrow \infty} r_n < \infty$. Set $\mathcal{P}_n = \{P \in \mathbb{Z}[X] \mid \deg P = n, h_B^p(P) \leq r_n\}$. It holds

1 $\lim_{n \rightarrow \infty} \frac{1}{\#\mathcal{P}_n} \sum_{P \in \mathcal{P}_n} |h_B^p(P) - h_{FS}(P) + \frac{1}{2}| = 0,$

2 $\lim_{n \rightarrow \infty} \frac{1}{\#\mathcal{P}_n} \sum_{P \in \mathcal{P}_n} \left| \frac{1}{n} \sum_{\substack{z \in \mathbb{C} \\ P(z)=0}} f(z) - \frac{i}{2\pi} \int_{\mathbb{C}} f(z) \frac{dz \wedge d\bar{z}}{(|z|^2+1)^2} \right| = 0.$

Proof of (a) (and of “(b1) $\Rightarrow$ (b2)”)

After rescaling the P_n 's we may assume $\|s_{P_n}\|_{\sup} = \sup_{z \in \mathbb{P}_{\mathbb{C}}^1} |s_{P_n}(z)|_{\text{FS}} = 1$.

$$\begin{aligned}
 & \left| \frac{1}{n} \sum_{\substack{z \in \mathbb{C} \\ P_n(z)=0}} f(z) - \frac{i}{2\pi} \int_{\mathbb{C}} f(z) \frac{dz \wedge d\bar{z}}{(|z|^2+1)^2} \right| = \left| \frac{1}{n} \int_{\text{div}(s_{P_n})} f(z) - \int_{\mathbb{P}_{\mathbb{C}}^1} f(z) c_1(\mathcal{O}(1)) \right| \\
 &= \frac{1}{n} \left| \int_{\mathbb{P}_{\mathbb{C}}^1} f(z) (\delta_{\text{div}(s_{P_n})} - c_1(\mathcal{O}(n))) \right| = \frac{1}{n} \left| \int_{\mathbb{P}_{\mathbb{C}}^1} f(z) \frac{i}{2\pi} \partial \bar{\partial} \log |s_{P_n}|_{\text{FS}} \right| \\
 &= \frac{1}{2\pi n} \left| \int_{\mathbb{P}_{\mathbb{C}}^1} \log |s_{P_n}|_{\text{FS}} \partial \bar{\partial} f \right| \leq \frac{1}{2\pi n} \int_{\mathbb{P}_{\mathbb{C}}^1} |\log |s_{P_n}|_{\text{FS}}| \cdot |\partial \bar{\partial} f| \\
 &\leq \frac{c_f}{n} \int_{\mathbb{P}_{\mathbb{C}}^1} -\log |s_{P_n}|_{\text{FS}} c_1(\mathcal{O}(1)) = c_f \left(\frac{1}{n} \log \|s_{P_n}\|_{\sup} - h_{\text{FS}}(P_n) + \frac{1}{2} \right) \\
 &\leq c_f \left(h_{\text{B}}^p(P_n) - h_{\text{FS}}(P_n) + \frac{1}{2} \right) + \frac{c}{n} \cdot \log n \xrightarrow{n \rightarrow \infty} 0.
 \end{aligned}$$

Hermitian line bundles on arithmetic varieties

- $\mathcal{X}/\mathbb{Z}$ arithmetic variety:
 - Integral scheme,
 - projective, flat, of finite type and separated over $\mathrm{Spec}(\mathbb{Z})$,
 - $\mathcal{X}_{\mathbb{Q}}$ smooth.
 - Example: $\mathbb{P}_{\mathbb{Z}}^1$.
- $\overline{\mathcal{L}} = (\mathcal{L}, h)$ hermitian line bundle on $\mathcal{X}$:
 - $\mathcal{L}$ line bundle on $\mathcal{X}$,
 - $h = (h_x)_{x \in \mathcal{X}(\mathbb{C})}$ system of hermitian metrics h_x on the fibers $\mathcal{L}_x$,
 - $|s(x)|^2 = h_x(s(x), s(x))$ is smooth in x on $\mathcal{U}(\mathbb{C})$ for every $s \in H^0(\mathcal{U}, \mathcal{L})$.
 - Curvature form: $c_1(\overline{\mathcal{L}}) = \frac{\partial \bar{\partial}}{2\pi i} \log |s|^2$ locally where s is invertible.
 - Example: $\overline{\mathcal{O}(1)}(\epsilon)$ is $\mathcal{O}(1)$ on $\mathbb{P}_{\mathbb{Z}}^1$ equipped with the Fubini–Study metric multiplied by $e^{-\epsilon}$: $|s| = |s|_{\mathrm{FS}} \cdot e^{-\epsilon}$.
- Small sections: A section s is called *small* (respectively *strictly small*) if $\|s\|_{\mathrm{sup}} = \sup_{x \in \mathcal{X}(\mathbb{C})} |s(x)| \leq 1$ (respectively $\|s\|_{\mathrm{sup}} < 1$).
- The small sections form a finite set

$$\widehat{H}^0(\mathcal{X}, \overline{\mathcal{L}}) = \{s \in H^0(\mathcal{X}, \mathcal{L}) \mid \|s\|_{\mathrm{sup}} \leq 1\}.$$

Equidistribution for arithmetic varieties

- $\overline{\mathcal{L}}$ is *arithmetically ample* if
 - $\mathcal{L}$ is relatively ample on $\mathcal{X}/\mathbb{Z}$,
 - $c_1(\overline{\mathcal{L}}) > 0$,
 - for $p \gg 0$ the set $H^0(\mathcal{X}, \mathcal{L}^{\otimes p})$ is generated by strictly small sections.
 - Example: $\overline{\mathcal{O}(1)}(\epsilon)$ is arithmetically ample for every $\epsilon > 0$:
 - (i) $\mathcal{O}(1)$ is ample on $\mathbb{P}_{\mathbb{Z}}^1$,
 - (ii) $c_1(\overline{\mathcal{O}(1)}(\epsilon)) = c_1(\mathcal{O}(1)) > 0$,
 - (iii) $\|Z_0^j Z_1^{p-j}\|_{\sup}^2 = e^{-2\epsilon p} \sup_{[z_0:z_1] \in \mathbb{P}_{\mathbb{Z}}^1(\mathbb{C})} \frac{|z_0^j z_1^{p-j}|^2}{(|z_0|^2 + |z_1|^2)^p} \leq e^{-2\epsilon p} < 1$.

Theorem (W. 2022)

Let $\mathcal{X}$ be an arithmetic variety of dimension $d = \dim \mathcal{X} \geq 2$ and $\overline{\mathcal{L}}$ an arithmetically ample hermitian line bundle on $\mathcal{X}$. For every $(d-2, d-2)$ C^0 -form Φ on $\mathcal{X}(\mathbb{C})$ it holds

$$\lim_{p \rightarrow \infty} \frac{1}{\#\widehat{H}^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes p})} \sum_{s \in \widehat{H}^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes p}) \setminus \{0\}} \left| \frac{1}{p} \int_{\text{div}(s)(\mathbb{C})} \Phi - \int_{\mathcal{X}(\mathbb{C})} \Phi \wedge c_1(\overline{\mathcal{L}}) \right| = 0.$$

Overview of the proof

- 1 Get rid of the form Φ by Stokes' theorem.
- 2 Equidistribution for random holomorphic sections in complex analysis by Bayraktar–Coman–Marinescu with respect to certain (non-discrete) probability measures on $H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})$.
- 3 Equidistribution for random sections in the real subspaces $H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$ of $H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{C}} \cong H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})$ for the probability measures supported on symmetric, compact and convex subsets $K_p \subseteq H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$ (e.g. $K_p = \{s \in H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}} \mid \|s\|_{\sup} \leq 1\}$).
- 4 Decompose $H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$ into boxes associated to the lattice points of $H^0(\mathcal{X}, \mathcal{L}^{\otimes p}) \subseteq H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$. By Moriwaki one can choose these boxes, such that they get exponentially fast small for $p \rightarrow \infty$.
- 5 Show: If $\lim_{p \rightarrow \infty} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s_p|| c_1(\overline{\mathcal{L}})^{d-1} = 0$ for a sequence of sections $s_p \in H^0(\mathcal{X}, \mathcal{L}^{\otimes p})$ with $\|s_p\|_{\sup} \leq 1$, then the same holds true for a sequence s'_p with $\limsup_{p \rightarrow \infty} \|s_p - s'_p\|^{1/p} < 1$.
- 6 Reduce from “almost all” sequences in $K_p \cap H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$ to “almost all” sequences in $K_p \cap H^0(\mathcal{X}, \mathcal{L}^{\otimes p}) (= \hat{H}^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes p}))$ by (4) and (5).

1. Get rid of the form Φ by Stokes' theorem

- By Stokes' theorem and the positivity of $c_1(\overline{\mathcal{L}})$ we can compute

$$\begin{aligned} \left| \frac{1}{p} \int_{\operatorname{div}(\mathbf{s})(\mathbb{C})} \Phi - \int_{\mathcal{X}(\mathbb{C})} \Phi \wedge c_1(\overline{\mathcal{L}}) \right| &= \frac{1}{p} \left| \int_{\mathcal{X}(\mathbb{C})} \Phi \wedge \frac{\partial \bar{\partial}}{2\pi i} \log |\mathbf{s}| \right| \\ &= \frac{1}{p} \left| \int_{\mathcal{X}(\mathbb{C})} \log |\mathbf{s}| \frac{\partial \bar{\partial}}{2\pi i} \Phi \right| \leq \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |\mathbf{s}|| \cdot \left| \frac{\partial \bar{\partial}}{2\pi i} \Phi \right| \\ &\leq \frac{c(\overline{\mathcal{L}}, \Phi)}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |\mathbf{s}|| c_1(\overline{\mathcal{L}})^{d-1} \end{aligned}$$

- Thus, it is enough to prove

$$\lim_{p \rightarrow \infty} \frac{1}{\#\widehat{H}^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes p})} \sum_{s \in \widehat{H}^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes p}) \setminus \{0\}} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |\mathbf{s}|| c_1(\overline{\mathcal{L}})^{d-1} = 0.$$

2. Equidistribution in complex analysis

Bayraktar, Coman and Marinescu introduced the following condition on probability measures and proved the following theorem:

Condition B

A sequence $(\sigma_p)_{p \geq 1}$ of probability measures on $H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})$ satisfies (B) if for all $p \geq 1$ there exists a constant $C_p \in \mathbb{R}$ such that

$$\int_{H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})} |\log |\langle s, u \rangle|| d\sigma_p(s) \leq C_p \quad \forall u \text{ with } \|u\| = 1,$$

where $H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})$ is equipped with the L^2 -product induced by the metric on $\bar{\mathcal{L}}$.

Theorem (BCM 2020)

If $(\sigma_p)_{p \geq 1}$ satisfies (B) with $\lim_{p \rightarrow \infty} \frac{C_p}{p} = 0$, then

$$\lim_{p \rightarrow \infty} \int_{H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s|| c_1(\bar{\mathcal{L}})^{d-1} d\sigma_p(s) = 0.$$

3. Equidistribution for real subspaces

- $(K_p)_{p \geq 1}$: Sequence of symmetric, compact and convex subsets

$$K_p \subseteq H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}} \subseteq H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{C}} \cong H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p}).$$

- Assume that the K_p 's are bounded by balls $\overline{B_{r'_p}} \subseteq K_p \subseteq \overline{B_{r_p}}$ with

$$\lim_{p \rightarrow \infty} r'_p{}^{1/p} = \lim_{p \rightarrow \infty} r_p{}^{1/p} = 1.$$

- Example: $K_p = \{s \in H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}} \mid \|s\|_{\text{sup}} \leq 1\}$.
- One can show, that the probability measures

$$\sigma_p = \frac{1}{\text{Vol}(K_p)} \lambda|_{K_p}$$

satisfy (B) with $\lim_{p \rightarrow \infty} \frac{C_p}{p} = 0$ for the Lebesgue measure λ on $H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$.

- We get

$$\lim_{p \rightarrow \infty} \frac{1}{\text{Vol}(K_p)} \int_{K_p} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s|| c_1(\overline{\mathcal{L}})^{d-1} d\lambda(s) = 0.$$

4. Decomposition into boxes at lattice points

- $H^0(\mathcal{X}, \mathcal{L}^{\otimes p}) \subseteq H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}}$ is a lattice.
- $s_{p,1}, \dots, s_{p,d_p} \in H^0(\mathcal{X}, \mathcal{L}^{\otimes p})$: Lattice basis of minimal norms.
- By Moriwaki there exists $\nu < 1$ with

$$\max_{j=1}^{d_p} \|s_{p,j}\| \leq C \cdot p^{d-1} \cdot \nu^p.$$

- For every $x \in H^0(\mathcal{X}, \mathcal{L}^{\otimes p})$ we define the box

$$Q_x = \left\{ x + \sum_{j=1}^{d_p} a_j s_{p,j} \mid a_j \in [0, 1) \right\} \cong [0, 1)^{d_p}.$$

- We get the disjoint decomposition

$$H^0(\mathcal{X}, \mathcal{L}^{\otimes p})_{\mathbb{R}} = \bigsqcup_{x \in H^0(\mathcal{X}, \mathcal{L}^{\otimes p})} Q_x.$$

5. Two near sequences of sections

- $(s_p)_{p \geq 1}, (s'_p)_{p \geq 1}$: Two sequences of sections $s_p, s'_p \in H^0(\mathcal{X}(\mathbb{C}), \mathcal{L}(\mathbb{C})^{\otimes p})$
- Assume:
 - i $\|s_p\|_{\sup} \leq 1,$
 - ii $\limsup_{p \rightarrow \infty} \|s_p - s'_p\|^{1/p} < 1,$
 - iii $\lim_{p \rightarrow \infty} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s_p|| c_1(\overline{\mathcal{L}})^{d-1} = 0.$
- Then one shows that it also holds

$$\lim_{p \rightarrow \infty} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s'_p|| c_1(\overline{\mathcal{L}})^{d-1} = 0.$$

- Idea of proof:
 - Prove it on smooth projective complex varieties by induction on $\dim$.
 - The case $\dim = 0$ is easy.
 - For the induction step use Stokes' theorem to reduce to the divisor of a section of a very ample line bundle.
 - By Bertini's theorem this divisor is in general smooth.
 - One gets an error term, which can be made smaller than any $\epsilon > 0$ by choosing the section using BCM 2020 (see (2.)).

6. Concluding by geometry of numbers

- $T_p \subseteq H^0(\mathcal{X}, \mathcal{L}^{\otimes p}) \cap K_p$: Subsets of positive density, that is

$$\liminf_{p \rightarrow \infty} \frac{\#T_p}{\#H^0(\mathcal{X}, \mathcal{L}^{\otimes p}) \cap K_p} > 0.$$

- It follows, that $\liminf_{p \rightarrow \infty} \frac{\text{Vol}(\bigsqcup_{x \in T_p} Q_x \cap K_p)}{\text{Vol}(K_p)} > 0$. By geometry of numbers one shows that the boxes intersecting ∂K_p do not matter.
- By (3.) there is a sequence $(s_p)_{p \geq 1}$ with $s_p \in \bigsqcup_{x \in T_p} Q_x \cap K_p$ such that

$$\lim_{p \rightarrow \infty} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s_p|| c_1(\bar{\mathcal{L}})^{d-1} = 0.$$

- By (4.) and (5.) there is a sequence $(s'_p)_{p \geq 1}$ with $s'_p \in T_p$ such that

$$\lim_{p \rightarrow \infty} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s'_p|| c_1(\bar{\mathcal{L}})^{d-1} = 0.$$

- As $\frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s'_p|| c_1(\bar{\mathcal{L}})^{d-1} \leq \widehat{\deg}(\bar{\mathcal{L}}^d)$ is bounded, we get

$$\lim_{p \rightarrow \infty} \frac{1}{\#K_p \cap H^0(\mathcal{X}, \mathcal{L}^{\otimes p})} \sum_{s \in K_p \cap H^0(\mathcal{X}, \mathcal{L}^{\otimes p}) \setminus \{0\}} \frac{1}{p} \int_{\mathcal{X}(\mathbb{C})} |\log |s|| c_1(\bar{\mathcal{L}})^{d-1} = 0.$$

Thank you for your attention!